Note of gluon PDFs

I. GLUON PDFS IN LATTICE

Arised from the so called "proton spin crisis", the Jaffe-Manohar decomposition [1] provides one such spin sum rule,

$$J = \frac{1}{2}\Delta\Sigma + L_q + L_G + \Delta G, \tag{1}$$

where $\frac{1}{2}\Delta\Sigma$ is the quark spin contribution, L_q and L_G are quark and gluon orbital angular momenta, and ΔG is the gluon spin contribution, which is found that the quark spin contribution to the proton spin is very small.

One can access ΔG via the first moment of the gluon helicity distribution $\Delta g(x)$,

$$\Delta G = \int dx \frac{i}{2xP^+} \int \frac{d\xi^-}{2\pi} e^{-ixP^+\xi^-} \langle PS| F_a^{+\alpha}(\xi^-) W^{ab}(\xi^-, 0) \tilde{F}^+_{\alpha,b}(0) |PS\rangle , \tag{2}$$

where $\xi^\pm = (\xi^0 \pm \xi^3)/\sqrt{2}$ are the light front coordinates and $W^{ab}(\xi^-, 0)$ is light-cone gauge-link in the adjoint representation. The gauge field tensor $F^{\mu\nu}$ and its dual $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$ are connected by the gauge link to construct a gauge invariant operator.

A. Distillation

Distillation [2] provides an optimized quark-smearing approach through low-rank approximation of the Jacobi smearing operator [3]. The latter implements gauge-covariant smearing via repeated applications of a lattice regularization of the 3D gauge-covariant Laplacian, are covariant derivatives. This spectral filtering suppresses short-distance fluctuations while amplifying physically relevant long-wavelength components of quark fields. The lattice Laplacian is identified as the simplest representation of the second-order three-dimensional differential operator

$$-\nabla^2_{xy}(t) = 6\delta_{xy} - \sum_{j=1}^{3} \left(U_j(x,t)\delta_{x+\hat{j},y} + U_j^\dagger(x-\hat{j},t)\delta_{x-\hat{j},y}\right), \tag{3}$$

where x and y correspond to the spatial coordinates in three-dimensional space. For a given time t, one can construct and solve the eigenvalue problem for the Laplace operator, obtaining eigenvalues $\lambda_1, ..., \lambda_M$ and their corresponding normalized eigenvectors $v^{(1)}, ..., v^{(M)}$, where $M = N_x \times N_y \times N_z \times N_c$.

$$-\nabla^2(t)v^{(i)} = \lambda_i v^{(i)}, \quad v^{(i)}v^{(j)\dagger} = \delta_{ij}. \tag{4}$$

From the lattice Laplacian, the Jacobi smearing kernel can be written as

$$J_{\sigma,n_\sigma}(t) = (1 + \frac{\sigma\nabla^2(t)}{n_\sigma})^{n_\sigma}, \quad \lim_{n_\sigma\to\infty} J_{\sigma,n_\sigma}(t) = \exp(\sigma\nabla^2(t)). \tag{5}$$

This implies that higher eigenmodes are exponentially suppressed, demonstrating that the dominant contributions originate from only a small subset of the lowest-magnitude modes. Consequently, the smearing kernel can be efficiently approximated by constructing a truncated eigenvector representation restricted to these low-lying modes. The distillation technique [2] is explicitly constructed using the lowest eigenmodes of the Jacobi kernel. Specifically, the distillation operator at time slice t is defined as

$$\Box_{xy}(t) = \sum_{k=1}^{N_{\rm ev}} v_x^{(k)}(t)\, v_y^{(k)\dagger}(t) \equiv V(t)V^\dagger(t), \tag{6}$$

where $V(t)$ is a $M \times N_{\rm ev}$ matrix and $N_{\rm ev}$ is the dimension of the distillation space. $v_x^{(k)}$ is the $k^{\rm th}$ eigenvector of ∇^2 at time slice t. The distillation operator is applied onto each quark field to achieve the smearing. The smeared annihilation operator for a nucleon can be written as

$$\begin{aligned}\mathcal{O}(t) &= \epsilon_{abc}(P_\pm)_\alpha\,(\Box\, u)^a_\alpha\left[(\Box\, u)^b_\beta\,(C\gamma_5)_{\beta\rho}\,(\Box\, d)^c_\rho\right],\\ &= \epsilon_{abc}(P_\pm)_\alpha\big(V_a^{(p)}V_b^{(q)}V_c^{(r)}(t)\big)\,\big(u_\alpha^{(p)}\left[u_\beta^{(q)}\,(C\gamma_5)_{\beta\rho}\,d_\rho^{(r)}\right]\big)\,\big(V^{(p)\dagger}V^{(q)\dagger}V^{(r)\dagger}(t)\big),\end{aligned} \tag{7}$$

where a,b,c are the color indices, p,q,r are the distillation indices and repeated spin indices α,β,ρ are summed. After integrating over the quark fields, the two-point correlator becomes

$$\begin{aligned}\langle\mathcal{O}(t)\bar{\mathcal{O}}(t_0)\rangle = (P_\pm)_{\alpha,\bar\alpha}&(C\gamma_5)_{\beta\rho}(C\gamma_5)_{\bar\beta\bar\rho}\\ &\times \epsilon_{abc}\big(V_a^{(p)}V_b^{(q)}V_c^{(r)}(t)\big)\\ &\times \big(V^{(p)\dagger}V^{(q)\dagger}V^{(r)\dagger}(t)\big)\,\mathcal{M}^{-1\,r,\bar r}_{\rho,\bar\rho}\left[\mathcal{M}^{-1\,p,\bar p}_{\alpha,\bar\alpha}\mathcal{M}^{-1\,q,\bar q}_{\beta,\bar\beta} - \mathcal{M}^{-1\,p,\bar q}_{\alpha,\bar\beta}\mathcal{M}^{-1\,q,\bar p}_{\beta,\bar\alpha}\right]\big(V^{(\bar p)}V^{(\bar q)}V^{(\bar r)}(t_0)\big)\\ &\times \big(V_{\bar a}^{(\bar p)\dagger}V_{\bar b}^{(\bar q)\dagger}V_{\bar c}^{(\bar r)\dagger}(t_0)\big)\,\epsilon_{\bar a\bar b\bar c},\end{aligned} \tag{8}$$

where $\mathcal{M}^{-1}$ is the lattice Dirac operator. The perambulators are defined as

$$\tau^{(p,\bar p)}_{\alpha,\beta} = V^{(p)\dagger}\,\mathcal{M}^{-1,p,\bar p}_{\alpha,\beta}\,V^{(\bar p)}, \tag{9}$$

the two-point correlator

$$\begin{aligned}\langle\mathcal{O}(t)\bar{\mathcal{O}}(t_0)\rangle = (P_\pm)_{\alpha,\bar\alpha}&(C\gamma_5)_{\beta\rho}(C\gamma_5)_{\bar\beta\bar\rho}\\ &\times \epsilon_{abc}\big(V_a^{(p)}V_b^{(q)}V_c^{(r)}(t)\big)\\ &\times \tau^{(r,\bar r)}_{\rho,\bar\rho}\,\big(\tau^{(p,\bar p)}_{\alpha,\bar\alpha}\tau^{(q,\bar q)}_{\beta,\bar\beta} - \tau^{(p,\bar q)}_{\alpha,\bar\beta}\tau^{(q,\bar p)}_{\beta,\bar\alpha}\big)\\ &\times \big(V_{\bar a}^{(\bar p)\dagger}V_{\bar b}^{(\bar q)\dagger}V_{\bar c}^{(\bar r)\dagger}(t_0)\big)\,\epsilon_{\bar a\bar b\bar c},\end{aligned} \tag{10}$$

The computationally intensive parallel transporters of the theory, known as perambulators, are solely dependent on the gauge field configuration and independent of the choice of interpolators. This key property enables the perambulators to be computed just once for each gauge field ensemble, after which they can be efficiently reused across an extended basis of interpolating operators. Such reuse provides significant computational savings, substantially reducing the overall numerical cost of calculations.

B. The field strength tensor

With the link variables U_μ, which are related to the A_μ fields $U_\mu(n) = \exp(iaA_\mu(n))$, the only gauge invariant object is the trace of a path ordered closed loop called Wilson loop. The simplest case is the plaquette, with the BakerCampbell-Hausdorff formula, which can be expressed as

$$P_{\mu\nu} = U_\mu(x)U_\nu(x+\hat\mu)U^\dagger_\mu(x+\hat\nu)U^\dagger_\nu(x) \tag{11}$$

$$= \exp\big(iaA_\mu(x) + iaA_\nu(x+\hat\mu) - iaA_\mu(x+\hat\nu) - iaA_\nu(x) - \frac{a^2}{2}[A_\mu(x), A_\nu(x+\hat\mu)] - \frac{a^2}{2}[A_\mu(x+\hat\nu), A_\nu(x)] \tag{12}$$

$$+\frac{a^2}{2}[A_\mu(x), A_\mu(x+\hat{\nu})] + \frac{a^2}{2}[A_\mu(x), A_\nu(x)] + \frac{a^2}{2}[A_\nu(x+\hat{\mu}), A_\mu(x+\hat{\nu})] + \frac{a^2}{2}[A_\nu(x+\hat{\mu}), A_\nu(x)] + \mathcal{O}(a^3)\big), \tag{13}$$

where $A_\mu(x+\hat{\nu}) = A_\mu(x) + a\partial_\nu A_\mu(x) + \mathcal{O}(a^2)$. With the Taylor expansion, it can be expressed as

$$P_{\mu\nu} = \exp\left(ia^2(\partial_\mu A_\nu(x) - \partial_\nu A_\mu(x) + i[A_\mu(x), A_\nu(x)]) + \mathcal{O}(a^3)\right) \tag{14}$$
$$= 1 + ia^2 F_{\mu\nu}(x) + \cdots, \tag{15}$$

where the field strength tensor $F_{\mu\nu}(x) = -i[D_\mu(x), D_\mu(x)]$ and $D_\mu(x) = \partial_\mu + iA_\mu(x)$. To reduce statistical fluctuations, the average of the four possible plaquettes that can be constructed by changing the signs of μ and ν is taken. This term is usually called the clover term, and the filed strength tensor can be expressed as

$$Q_{\mu\nu}(x) = P_{\mu\nu}(x) - P_{\nu-\mu}(x) + P_{-\nu\mu}(x) + P_{-\mu-\nu}, \tag{16}$$
$$F_{\mu\nu} = -\frac{i}{8a^2}(Q_{\mu\nu} - Q_{\nu\mu}), \tag{17}$$

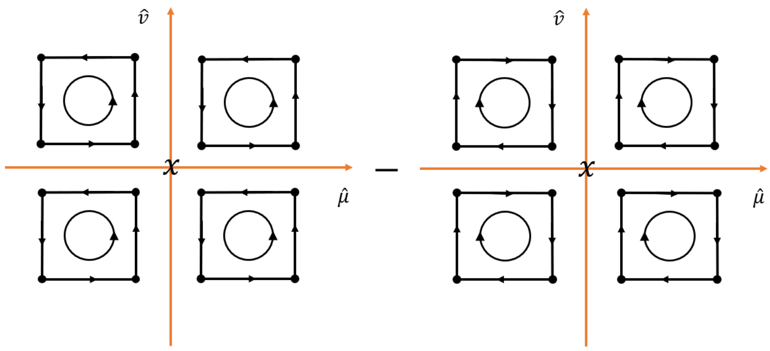

FIG. 1: The clover term and the way $F_{\mu\nu}$ discretized.

In the Minkowskian space, the covariant and the contravariant field strength tensor can be written

$$F^{(\mathbf{M})}_{\mu\nu} = \begin{pmatrix} 0 & F_{tx} & F_{ty} & F_{tz} \\ F_{xt} & 0 & F_{xy} & F_{xz} \\ F_{yt} & F_{yx} & 0 & F_{yz} \\ F_{zt} & F_{zx} & F_{zy} & 0 \end{pmatrix}, \quad F^{\mu\nu,(\mathbf{M})} = g^{\mu\rho,(\mathbf{M})} F^{(\mathbf{M})}_{\rho\sigma} g^{\sigma\nu,(\mathbf{M})} = \begin{pmatrix} 0 & -F_{tx} & -F_{ty} & -F_{tz} \\ -F_{xt} & 0 & F_{xy} & F_{xz} \\ -F_{yt} & F_{yx} & 0 & F_{yz} \\ -F_{zt} & F_{zx} & F_{zy} & 0 \end{pmatrix}, \tag{18}$$

The relations of tx, ty, xy components between the Minkowskian and the Euclidean quantities canc be written as

$$F^{tx,(\mathbf{M})} F^{(\mathbf{M})}_{tx} = -F^{(\mathbf{E})}_{tx} F^{(\mathbf{E})}_{tx}, \quad F^{xy,(\mathbf{M})} F^{(\mathbf{M})}_{xy} = F^{(\mathbf{E})}_{xy} F^{(\mathbf{E})}_{xy}, \tag{19}$$

C. The unpolarized gluon operator

The matrix elements for the unpolarized gluon PDF in the fundamental representation can be written as

$$M_{\mu\alpha;\lambda\beta}(z,p) = \langle p| F_{\mu\alpha}(z) W[z,0] F_{\lambda\beta}(0) W[0,z] |p\rangle. \tag{20}$$

where $z_\mu = \{0,0,0,z_3\}$ is the separation between the gluon fields. The matrix elements can be decomposed into the invariant amplitudes, $\mathcal{M}_{pp}, \mathcal{M}_{pz}, \mathcal{M}_{zz}, \mathcal{M}_{zp}, \mathcal{M}_{ppzz}$ and $\mathcal{M}_{gg}$ [4]. Since $g_{++} = g_{--} = 0, F_{++} = 0$, the light-cone gluon distribution is obtained from

$$g^{\alpha\beta}M_{+\alpha;\beta+}(z,p)=g^{ij}M_{+i;j+}(z,p)=-2p_+^2\mathcal{M}_{pp}(\nu,0), \tag{21}$$

where the Ioffe time $\nu=p_3z_3$. The gluon PDF is determined by the $\mathcal{M}_{pp}$ amplitude,

$$-\mathcal{M}_{pp}=\frac{1}{2}\int_{-1}^{1}dxe^{-ix\nu}xg(x). \tag{22}$$

The matrix elements $M_{ti;it}$ and $M_{ij;ji}$ can be combined to extract the twist-2 invariant amplitudes $\mathcal{M}_{pp}$ as

$$M_{ti;it}+M_{ji,ij}=2p_0^2\mathcal{M}_{pp}. \tag{23}$$

Both $M_{ti;it}$ and $M_{ij;ji}$ have the same one-loop UV anomalous dimension, making the whole combination in Eq. 23 multiplicatively renormalizable at the one-loop level, at least [4].

The gluonic currents, inserted into the nucleon to calculate the matrix elements, are not connected to the nucleon state by any quark propagator, so the currents are largely decoupled from the nucleon part of the calculation itself. As a result, on the lattice, the gluonic currents and the nucleon two-point correlators can be calculated separately. The three-point correlator can be rewritten as

$$C_{3pt}(z,P_z,t,t_i,t_0)=\langle(\mathcal{O}_g(z,t_i)-\langle\mathcal{O}_g(z,t_i)\rangle)(C^N_{2pt}(P_z,t,t_0)-\langle C^N_{2pt}(P_z,t,t_0)\rangle)\rangle\,, \tag{24}$$

where $\langle\cdots\rangle$ indicates the ensemble average. In the fundamental representation, the gluon current defined with the matrix elements in Eq. 23 can be written as

$$\begin{aligned}\mathcal{O}_g^{(0)}(z,t_i)&=\big(F^{0i}(z,t_i)W[z,0]F_{0i}(0,t_i)W[0,z]+2F^{12}(z,t_i)W[z,0]F_{12}(0,t_i)W[0,z]\big)_{\mathbf{Mink}}\\&=\big(-F^{0i}(z,t_i)W[z,0]F_{0i}(0,t_i)W[0,z]+2F^{12}(z,t_i)W[z,0]F_{12}(0,t_i)W[0,z]\big)_{\mathbf{Eucl}}.\end{aligned} \tag{25}$$

The "-" sign for $F_{0i}(z,t_i)W[z,0]F_{i0}(0,t_i)W[0,z]$ term is introduced for the relations between the Minkowskian and the Euclidean quantities in Eq. 19. In addition, to investigate the contribution of $M_{ti;it}$, we also selected another operator in Euclidean space

$$\mathcal{O}_g^{(1)}(z,t_i)=F^{0i}(z,t_i)W[z,0]F_{0i}(0,t_i)W[0,z]. \tag{26}$$

The two-point functions

$$C^N_{2pt}(P_z,t,t_0)=\mathcal{P}_{\alpha\beta}\sum_{\vec{x},\vec{y}}e^{-iP_zz}\left\langle 0\right|N_\alpha(\vec{x},t)\bar{N}_\beta(\vec{y},t_0)\left|0\right\rangle, \tag{27}$$

where the projector $\mathcal{P}=(1\pm\gamma_0)/2$.

The gauge configurations used in this work are generated by CLQCD collaboration with $N_f=2+1$ Wilson Clover quark action. The detailed information about the ensembles are listed in Table. I. We consider the configurations with ten steps of hypercubic smearing (HYP5), and we compared them with that applied by Wilson flow with flow time $\tau=1.4$.

TABLE I: The parameters of the ensemble used in this work.

ID	a(fm)	β	m_π	$L^3\times N_t$	N_{conf}
C11P29S	0.105	6.20	290	$24^3\times 72$	317

The distillation quark smearing method is used to compute the quark propagators. A phase is added to the distillation eigenvectors for the higher momenta preserving the translational invariance, which is essential for the projection onto the states of definite momenta [5]. The phased distillation eigenvector and the momentum smearing phase factor is chosen as

$$\tilde{v}_x^k(\vec{z},t) = e^{i\vec{\zeta}\cdot\vec{z}} v_x^k(\vec{z},t), \quad \vec{\zeta} = 2 \times \frac{2\pi}{L}\hat{z}, \tag{28}$$

giving the momentum smearing needed for boosts up to $P = 6 \times \frac{2\pi}{aL}$. In Fig. 2 and Fig. 3, the ratio of e three-point function to the two-point function with different gluon current operator defined in Eq. 25 and Eq. 26 and $P_z = (0,0,2)$. The configurations in Fig. 2 and Fig. 3 are applied with HYP5 smearing. While in the Fig. 4 and Fig. 5, they are applied with Wilson3 smearing with flow time $\tau = 1.4$.

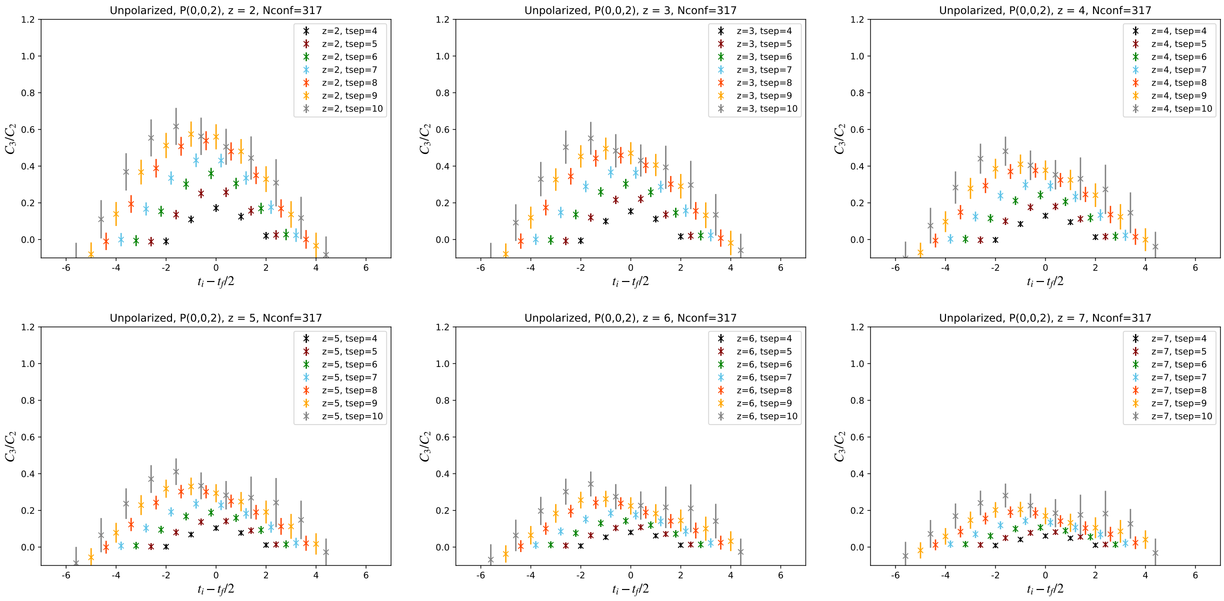

FIG. 2: Ratio of the three-point function to the two-point function with the unpolarized gluon current operator $\mathcal{O}_g^{(0)}$ defined in Eq. 25 and $P_z = (0,0,2)$. The configurations are applied with HYP5 smearing.

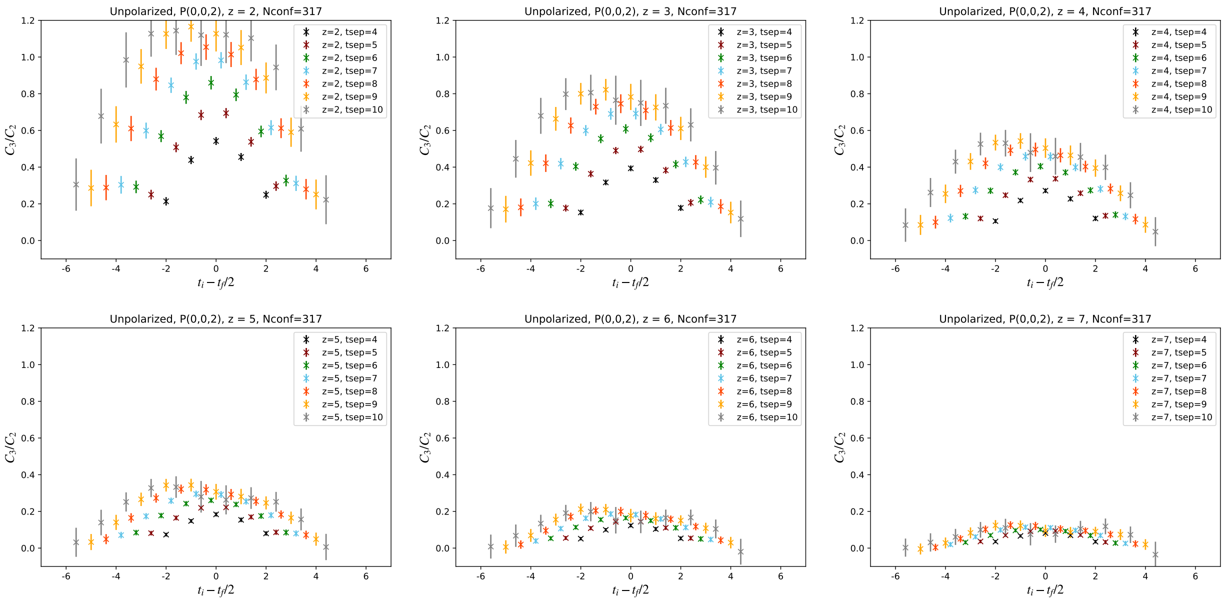

FIG. 3: Ratio of the three-point function to the two-point function with the unpolarized gluon current operator $\mathcal{O}_g^{(1)}$ defined in Eq. 26 and $P_z = (0, 0, 2)$. The configurations are applied with HYP5 smearing.

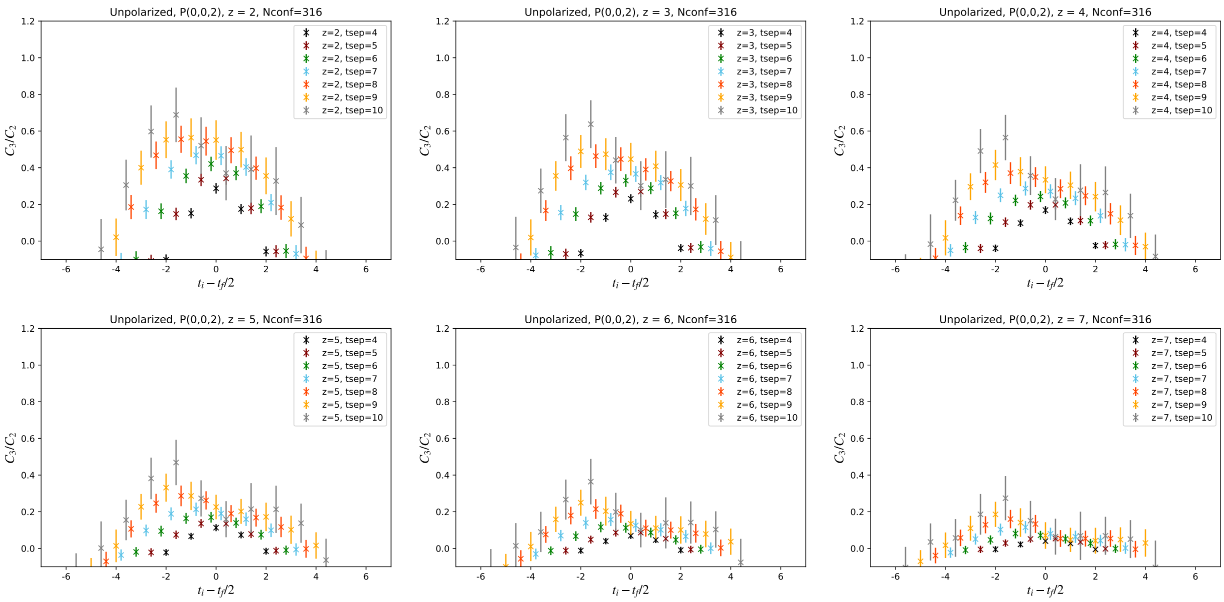

FIG. 4: Ratio of the three-point function to the two-point function with the unpolarized gluon current operator $\mathcal{O}_g^{(0)}$ defined in Eq. 25 and $P_z = (0, 0, 2)$. The configurations are applied with Wilson3 smearing.

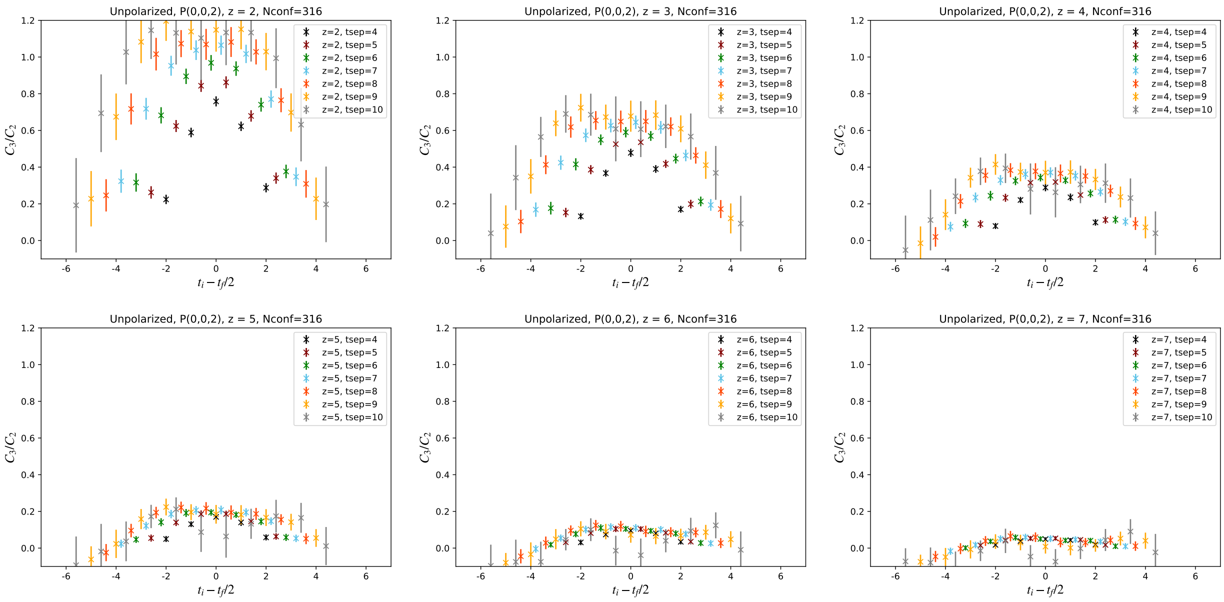

FIG. 5: Ratio of the three-point function to the two-point function with the unpolarized gluon current operator $\mathcal{O}_g^{(1)}$ defined in Eq. 26 and $P_z = (0, 0, 2)$. The configurations are applied with Wilson3 smearing.

D. The gluon helicity operator

In the fundamental representation, the gluon helicity matrix elements can be written as

$$\tilde{m}_{\mu\alpha;\lambda\beta}(z,p) = \langle ps| F_{\mu\alpha}(z)W[z,0]\tilde{F}_{\lambda\beta}(0)W[0,z] |ps\rangle \,, \tag{29}$$

where $z_\mu = \{0,0,0,z_3\}$ is the separation between the gluon fields. The z-odd combination of the matrix elements [6] is defined as,

$$\tilde{M}_{\mu\alpha;\lambda\beta}(z,p) = \tilde{m}_{\mu\alpha;\lambda\beta}(z,p) - \tilde{m}_{\mu\alpha;\lambda\beta}(-z,p). \tag{30}$$

The usual light-cone polarized gluon distribution Δg is obtained from the matrix element $g^{\alpha\beta}\tilde{M}_{+\alpha;\beta+}(z,p)$, and it can be reduces to $g^{ij}\tilde{M}_{+i;j+}(z,p) = \tilde{M}_{+i;+i}(z,p)$, since $g_{++} = g_{--} = 0, F_{++} = 0$. The matrix elements

$$\tilde{M}_{+i;+i} = \tilde{M}_{0i;0i} + \tilde{M}_{3i;3i} + \tilde{M}_{0i;3i} + \tilde{M}_{3i;0i}. \tag{31}$$

The polarized gluon PDF can be determined by the Ioffe-time distribution [7]

$$\Delta G = \int_0^\infty d\nu \tilde{\mathcal{I}}_p(\nu) = \int_0^1 dx \Delta g(x), \quad -i\tilde{\mathcal{I}}_p(\nu) \equiv \tilde{\mathcal{M}}^{(+)}_{(ps)}(\nu) - \nu\tilde{\mathcal{M}}_{(pp)}(\nu), \tag{32}$$

with the Ioffe time $\nu = p_3 z_3$. And the combination of the matrix elements $\tilde{M}_{0i;0i}$ and $\tilde{M}_{ij;ij}$ can be written in terms of invariant amplitudes as

$$\begin{aligned}\tilde{M}_{0i;0i}(z,p) + \tilde{M}_{ij;ij}(z,p) &= -2p_3p_0\tilde{\mathcal{M}}^{(+)}_{(ps)}(\nu,z^2) + 2p_0^3 z\tilde{\mathcal{M}}_{(pp)}(\nu,z^2) &(33)\\ &= -2p_3p_0[\tilde{\mathcal{M}}^{(+)}_{(ps)}(\nu,z^2) - \nu\tilde{\mathcal{M}}_{(pp)}(\nu,z^2) + \frac{m^2}{p_3^2}\nu\tilde{\mathcal{M}}_{(pp)}(\nu,z^2)]. &(34)\end{aligned}$$

This combination becomes proportional to the desired amplitude $\tilde{\mathcal{M}}^{(+)}_{(ps)}(\nu) - \nu\tilde{\mathcal{M}}_{(pp)}(\nu)$ in Eq. 32 for large p_3.

To extract the gluon helicity distribution, we calculate the combination $\tilde{M}_{0i;0i}(z,p) + \tilde{M}_{ij;ij}(z,p)$. And the gluon current operator,

$$\begin{aligned}\mathcal{O}^{(0)}_g(z,t_i) &= \big(\{F^{0i}(z,t_i)W[z,0]\tilde{F}_{0i}(0,t_i)W[0,z] + F^{ij}(z,t_i)W[z,0]\tilde{F}_{ij}(0,t_i)W[0,z]\} - \{(z\leftrightarrow -z)\}\big)_{\mathbf{Mink}}\\ &= -\big(\{F^{01}(z,t_i)W[z,0]F^{23}(0,t_i)W[0,z] - F^{02}(z,t_i)W[z,0]F^{13}(0,t_i)W[0,z]\\ &\quad + 2F^{12}(z,t_i)W[z,0]F^{03}(0,t_i)W[0,z]\} - \{(z\leftrightarrow -z)\}\big)_{\mathbf{Eucl}},\end{aligned} \tag{35}$$

where $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$ and $\epsilon^{0123} = 1$. Similarly to the unpolarized case, we defined another gluon current operator in Euclidean space

$$\begin{aligned}\mathcal{O}^{(1)}_g(z,t_i) &= \{F^{0i}(z,t_i)W[z,0]\tilde{F}_{0i}(0,t_i)W[0,z]\} - \{(z\leftrightarrow -z)\}\\ &= \{F^{01}(z,t_i)W[z,0]F^{23}(0,t_i)W[0,z] + F^{02}(z,t_i)W[z,0]F^{13}(0,t_i)W[0,z]\} - \{(z\leftrightarrow -z)\},\end{aligned} \tag{36}$$

The two-point functions

$$C^N_{2pt}(P_z,t,t_0) = \mathcal{P}_{\alpha\beta}\sum_{\vec{x},\vec{y}} e^{-iP_z z}\langle 0| N_\alpha(\vec{x},t)\bar{N}_\beta(\vec{y},t_0) |0\rangle \,, \tag{37}$$

For the unpolarized two-point functions, the projector $\mathcal{P} = (1\pm\gamma^0)/2$, and for the helicity case, the projector $\mathcal{P} = i\gamma^3\gamma^5(1\pm\gamma^0)/2$ in the Euclidean space. In Fig. 6 and Fig. 7, the ratio of e three-point function to the two-point function with different gluon current operator defined in Eq. 35 and Eq. 36 and $P_z = (0,0,2)$. The configurations in

Fig. 6 and Fig. 7 are applied with HYP5 smearing. While in the Fig. 8 and Fig. 9, they are applied with Wilson3 smearing with flow time $\tau = 1.4$.

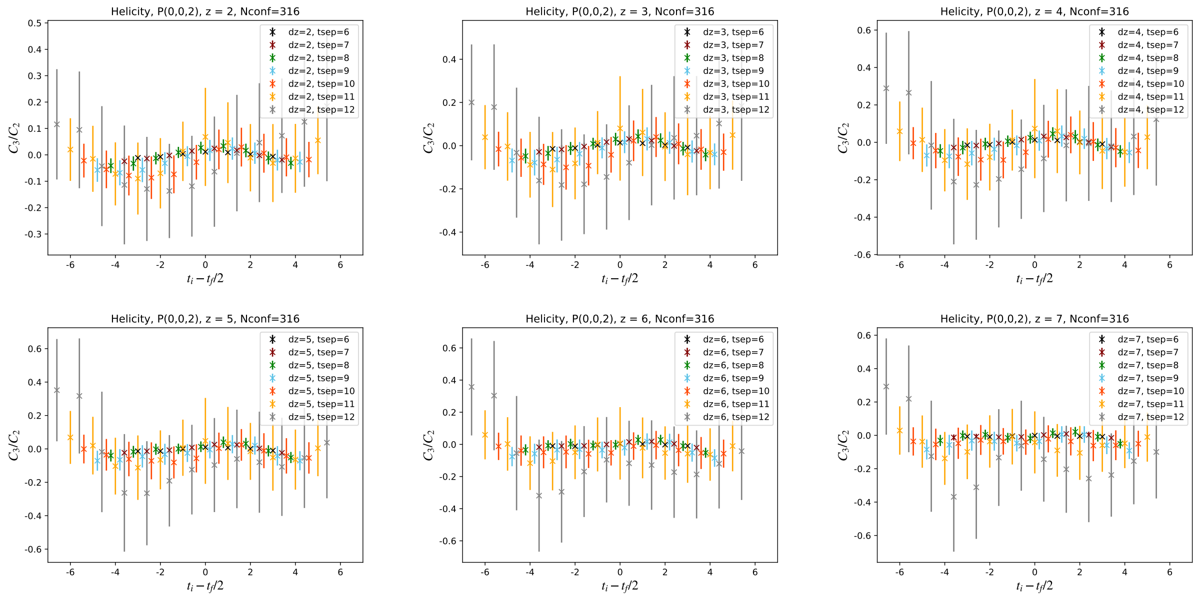

FIG. 6: Ratio of the three-point function to the two-point function with the helicity gluon current operator $\mathcal{O}_g^{(0)}$ defined in Eq. 35 and $P_z = (0, 0, 2)$. The configurations are applied with HYP5 smearing.

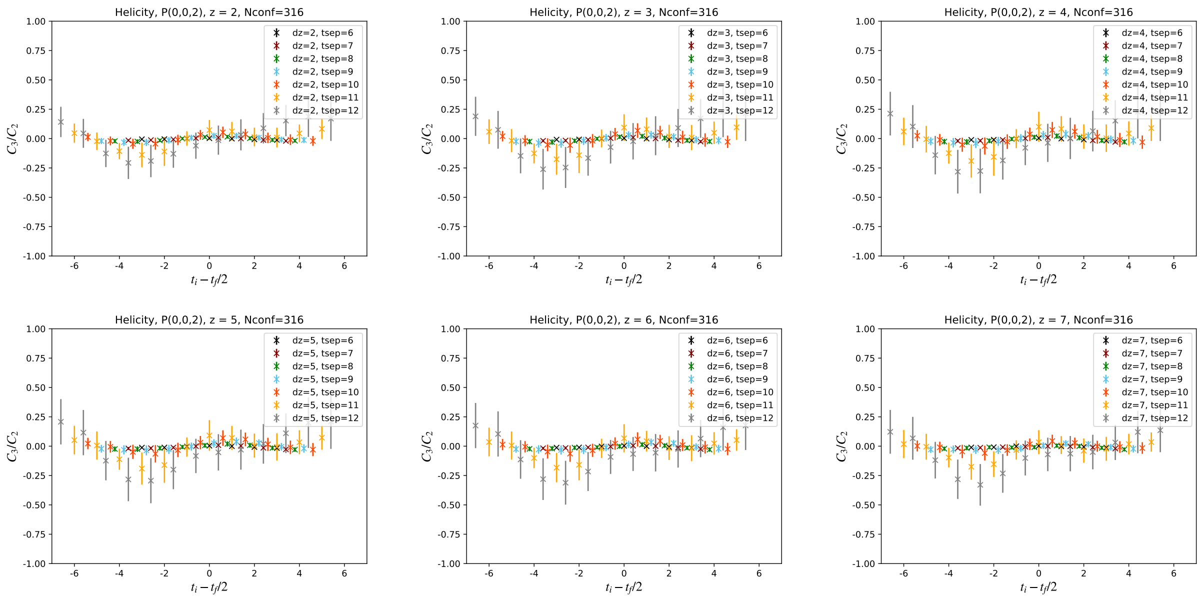

FIG. 7: Ratio of the three-point function to the two-point function with the helicity gluon current operator $\mathcal{O}_g^{(1)}$ defined in Eq. 36 and $P_z = (0, 0, 2)$. The configurations are applied with HYP5 smearing.

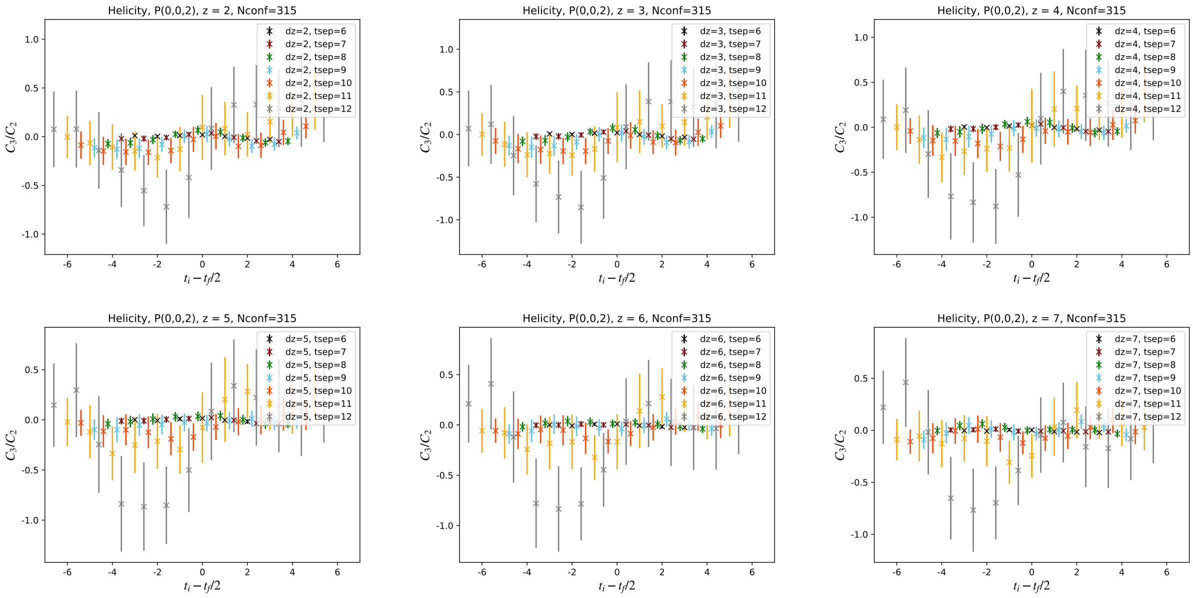

FIG. 8: Ratio of the three-point function to the two-point function with the helicity gluon current operator $\mathcal{O}_g^{(0)}$ defined in Eq. 35 and $P_z = (0, 0, 2)$. The configurations are applied with Wilson3 smearing.

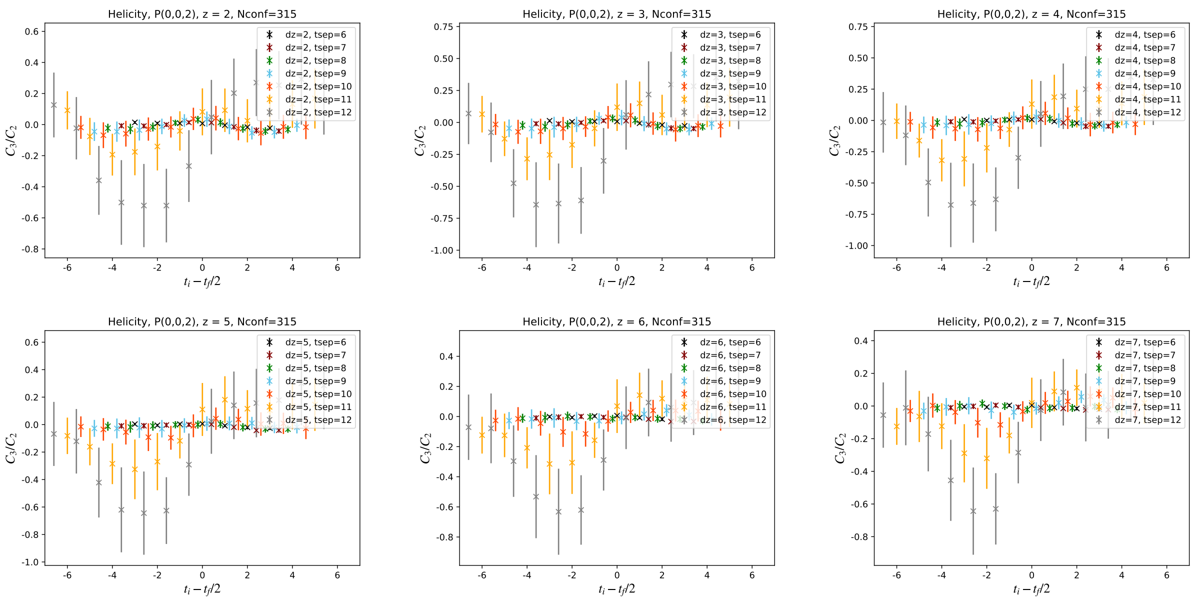

FIG. 9: Ratio of the three-point function to the two-point function with the helicity gluon current operator $\mathcal{O}_g^{(1)}$ defined in Eq. 36 and $P_z = (0, 0, 2)$. The configurations are applied with Wilson3 smearing.

E. The gluon helicity operator

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